Roll No. | | | | | | | |

Paper Code: MGI-313

M.Tech, Geo-Informatics

Application of Geo-Informatics for Natural resource management (NRM)

End Semester Examination 2025

III Semester

Time: 03:00Hour

Maximum Marks: 100

Note:

(i) Answer **all the questions** by choosing ***any two of the sub questions***.

(ii) Each question carries 10 marks

Q1.	Any two (20 Marks)	
(a)	Explain the concept of land suitability assessment with reference to agricultural planning and management.	CO1
(b)	Discuss the application of RS and GIS tools in analyzing geomorphological features and landform characteristics.	CO1
(c)	Illustrate how remote sensing technologies assist in the planning and implementation of watershed management programs	CO1
Q2.	**Any two** (20 Marks)	
(a)	Describe the use of microwave remote sensing in soil moisture estimation and its role in monitoring crop growth dynamics.	CO2
(b)	Define soil moisture and explain its significance in agricultural and environmental applications.	CO2
(c)	Analyze the importance of GIS-based integration of thematic layers for hazard mapping and zonation studies.	CO2
Q3.	**Any two** (20 Marks)	
(a)	Explain how satellite imagery contributes to detecting and analyzing forest cover changes over time.	CO3
(b)	Compare the spectral and wavelength characteristics of panchromatic, multispectral, and hyperspectral sensors.	CO3
(c)	Describe the essential steps and techniques involved in visual and digital image interpretation	CO3
Q4.	**Any two** (20 Marks)	
(a)	Demonstrate the utility of geospatial datasets in identifying and mapping biodiversity hotspots.	CO4
(b)	Explain the role of remote sensing data in assessing crop vigor and stress levels within the framework of precision agriculture.	CO4
(c)	Discuss the various image classification techniques, emphasizing their dependence on spectral properties and availability of ground truth information	CO4
Q5.	**Any two** (20 Marks)	
(a)	Discuss the significance of Synthetic Aperture Radar (SAR) in natural resource monitoring and management applications	CO5
(b)	Explain briefly how remote sensing techniques are utilized for forest cover mapping and long-term monitoring	CO5
(c)	Illustrate the process and importance of crop acreage estimation and inventory generation using geospatial technologies.	CO5